

Fires, Forests and Futures: The ANU Westoby Lecture

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Fires, Forests and Futures: The ANU Westoby Lecture

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Summary

This paper addresses management and policy issues arising from the 2009 and other recent fires that burnt the iconic ash-type forests of eastern Victoria. After sketching a general framework for an analysis of sustainability, the paper elaborates the results of a simulation study on seed collection and storage (to be published separately), especially in relation to the risks posed to the ash-type state forests by major fires. These forest types are notably fire sensitive and easily killed by fires of moderate or even mild intensity. They are also cyclical in the extent of seed production and do not set seed in commercial quantities for the first 20 years of life. If seed collection and storage is restricted to 'low' levels until 2050, the risks are substantial, involving probabilities over 90% for alpine ash, implying insufficient seed to permit artificial regeneration of those burnt areas that have not regenerated naturally, whether climate change occurs or not. Those risks can be reduced substantially by higher levels of seed collection, but even at 'median' levels the probabilities may approach 40% under high rates of climate change. The simulated outcomes for mountain ash state forests are much less severe but the impacts of high rates of climate change are still great enough to cause discomfort.

Data for the ash-type conservation reserves were not available but descriptive reports indicate that the risks for both forest types in the conservation reserves may be very high, as many of the old-growth stands in the national parks (often closed water catchments) were severely burnt in the 2003, 2006–2007 or 2009 fires. The risks posed by fire in the next 20 y to water quality and biodiversity are very likely to be material.

This leads to advocacy of active management measures that include: (1) a well-equipped workforce, (2) removal and salvage of fire-killed trees along access roads, (3) seed collection and artificial regeneration in conservation reserves, (4) strategic swaps of age classes to the advantage of both tenures, (5) regrowth thinning for habitat management and water production, (6) better access for fire management and more fuel reduction burning and (7) strategic orientation of harvest areas.

In the past, the conservation versus development debate about forests has been a 'zero-sum game'. Whatever one side gained the other lost, as areas were transferred from one designation to the other. That approach neglects the role of a third player,

Mother Nature, especially relating to fire, and this means that a joint approach to addressing the risks is desirable.

The ash-type conservation reserves are as much about multiple uses as are state forests. Some uses are common to both, some excluded from one and not the other, and some that vary by degree rather than absolute exclusion in one or the other. We face a future in which there is a material risk that we will lose most or all of these ash-type forests, not over millennia, but in the next 60 y. We can decide to do nothing and accept that loss, or we can engage in active management to conserve their values and uses by reducing the risks across both tenures.

Keywords: fire; risk; seed production; seed collection; forest management; planning; Monte Carlo method; *Eucalyptus regnans*; *Eucalyptus delegatensis*

Introduction

This lecture was dedicated to Jack Westoby, the former head of the FAO Forestry Division. As is well known, Jack underwent an epiphany late in his career, leading him to question the fundamental creed that he and his organisation had pursued in the years following World War II. He argued that forests were for people, not for their intrinsic worth, and captured the nub of environmental and sustainability concerns in his argument. Few who heard his 'Kangaroo' speech that used a quote from the D.H. Lawrence novel as an opener in the 1974 FORWOOD conference will forget the power of his argument.

Alf Leslie was a close colleague of Jack Westoby. In my view, he was as much responsible for this conversion as was Jack. I can recollect the discussions with Alf in Rome in which he was advancing similar views prior to Jack's epiphany. So this lecture is also dedicated to Alf, as he was a much-respected teacher, mentor, colleague, friend and sceptic over many years. Alf loved a contrary argument. While I could not emulate his approach or style, I tried to illuminate some current issues in dealing with 'fires, forests and futures' in the light of recent history in Victoria.

Sustainability

Much attention has rightly been focused on sustainability in recent years and on sustainable forest management in particular. The most

widely cited definition of sustainability rests on the definition of sustainable development by the Bruntland Commission (1997):

Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Sustainable forests and forestry

To an economist, the central theme of this definition amounts to a concern about intergenerational equity — how we balance the preference for present consumption against our concern for that of future generations. To some economists, but not all, the equity tag places it outside the direct province of economic theory.

Some time ago (Ferguson 1996), I tried to rationalise that desire for intergenerational equity in valuing utility over time through consideration of the social rate of time preference, the discount rate that measures the relative preference for present over future consumption. I posited a discount rate that followed commercial values over the investment time horizons that are common (i.e. generally less than 50 y), but then declined progressively thereafter until it reached a steady low state founded on our unwillingness to discriminate between the consumption of successive future generations in the long-distant future.

Argument over the social discount rate has a long history and is continuing. Koopmans (1960) showed that under certain assumptions characterising utility functions that the discount rate should be constant. Beltratti *et al.* (1994) introduced an approach that was based on achieving the highest indefinitely sustainable welfare — the so-called ‘Green Golden Rule’, in which the long-term future is the sole dictator of the social discount rate, which therefore resides at a constant rate of zero. Chichilnisky (1996) showed that with the addition of further rigorous assumptions about the long-term future, a time path for the social discount rate similar to the one I have described could be justified. Heal (1998) describes this latter approach as the balance between a ‘dictatorship of the present’ and a ‘dictatorship of the future’.

But if any one of these three starkly differing approaches is an appropriate device for valuing resource consumption issues over time, how come it is so little adopted? Economic studies that choose one or the other in analysing major issues such as climate change have not been widely accepted, and choice is left to the political process.

Part of the reason is that other factors mask whatever path the social discount rate takes. Skidelsky (2008), the biographer of Keynes, quotes him as describing economics as ‘a polite technique which tries to shield us from the fact that we know very little about the future’. Keynes (1936) was forthright in arguing that the long-term future is unpredictable and went on to underscore that point many times, for example, ‘The outstanding fact is the extreme precariousness of the basis of knowledge on which our estimates of prospective yield have to be made.’ Of course, a sceptic might extend his barb directed at economics to many other disciplines, including ecology, forestry and climate sciences.

We are stuck with the dilemma that we should look to a compromise between the dictatorship of the present and a dictatorship of the distant future, knowing that the immediate future is variable and

erratic, and the long-distant future is unpredictable. Central to this dilemma is determining, as a society, what values we are seeking to maintain in different forest ecosystems in both the immediate future and the longer term. Anticipating the needs of future generations, or assuming that they will value the same environmental attributes as does our generation, is a continuing challenge, not only for economics but for the political process as well.

To address this dilemma, three features were introduced in constructing a simulation model for analysis of futures in the ash-type forests of Victoria.

The first is that forest planning over time is best restricted to a planning horizon of 40–50 y, because attempting to forecast beyond that time is a futile exercise. That is not to say the long-distant future should be neglected. Rather, it forms the linchpin of the second feature.

The second feature is that we should focus on the state of the forest (i.e. the structural and spatial properties) at the end of planning horizon to ensure that we have left it in a better condition for those future generations, that is, a system that is capable of providing the things that we currently value, or that future generations might value. In other words, the terminal state at the end of the planning horizon becomes another constraint in the model. This sidesteps the issue of the longer-term path of the discount rate but also provides a sharper focus on the goals we are setting for the longer-distant future. Ideally, of course, the model should also be spatial in content, which this model is not. Nevertheless, we are slowly moving towards that goal as computing power and speed increases. This will enable the issues of variability and erratic changes due to natural disasters such as fire to be better addressed.

The third feature concerns recognition of the impact of natural disturbances, and fires in particular, on forest planning. These impacts are much broader than those of harvest scheduling or planning horizons, but the limit on the planning horizons and consideration of the condition of the forest at the end of the horizon are still appropriate. Indeed, such an approach can be applied to advantage to the broader forest estate that includes conservation reserves as well as the smaller areas of state forests now available for timber harvesting.

Ash-type forests

The prompt for the title of this lecture stemmed from a study of seed-storage requirements following the 2009 fires for the ash-type forests in Victoria, commissioned by the Department of Sustainability and Environment. Technical details of the study are reported in Ferguson (2010), so this lecture described and illustrated the relevant material in general terms and avoided detailed repetition of source material, methods, equations and results, except for those germane to the thesis of the lecture.

Much has been written about the ash-type forests of Victoria. Two major and distinct forest types are involved; the mountain ash (*Eucalyptus regnans*) forests and alpine ash (*E. delegatensis*) forests. There are smaller areas of related forest types such as shining gum (*E. nitens*), but to keep the presentation simple I

referred only to these two forest types and the dominant species that characterise them. The differentiation in altitude that the respective names imply results in a predominantly separate spatial geography, as shown in the Figure 1.

Mountain ash forests are the archetypal tall forests of Australia and are mainly concentrated in one large patch in the Central Highlands of Victoria. The patch is interspersed by higher-elevation ridges carrying alpine ash, snow gums and alpine meadows and by foothill ridges carrying mixed species such as messmate stringybark and peppermint. Alpine ash forests generally have a less complex and diverse understorey than mountain ash and are scattered more widely in patches across the Eastern Victoria highlands from the eastern end of the Central Highlands, through the Great Divide, and into the far eastern Alpine area — to use some names that recur in formal reports (Wareing and Flinn 2003; Flinn *et al.* 2009).

Ash-type forests are an iconic part of our natural heritage. They are aesthetically attractive, highly productive of market values such as wood and biomass, and provide important non-market catchment and biodiversity values and wildlife habitat. They are divided by formal tenure into conservation reserves (mainly

national parks) in which no directly exploitive uses are permitted, and state forests in which multiple uses including wood production and certain other exploitive uses are permitted.

Fires

Let us restrict our attention initially to state forests. Most of the mountain ash in state forests originated from a very extensive fire in 1939, but some smaller stands of regrowth from 1919 and 1926 fires were either not burnt or not much damaged by the 1939 fire. The 1939 Black Friday fire was a fire that by any measure was intense — possibly more intense in places than the recent 2009 Black Saturday fire. In 1939, about 85% of mountain ash forests now in state forests was burnt at such high intensities that the overwhelming majority of trees were killed. Fortunately that was a good seed year and in most places prolific regeneration followed on a seedbed in which competition from other plants had been much reduced by the intensity of the fire. There were exceptions, such as patches where earlier fires had produced regrowth that in 1939 was too young to bear seed, or where the seed crop was locally insufficient, resulting in dense regeneration of silver wattle or other species.

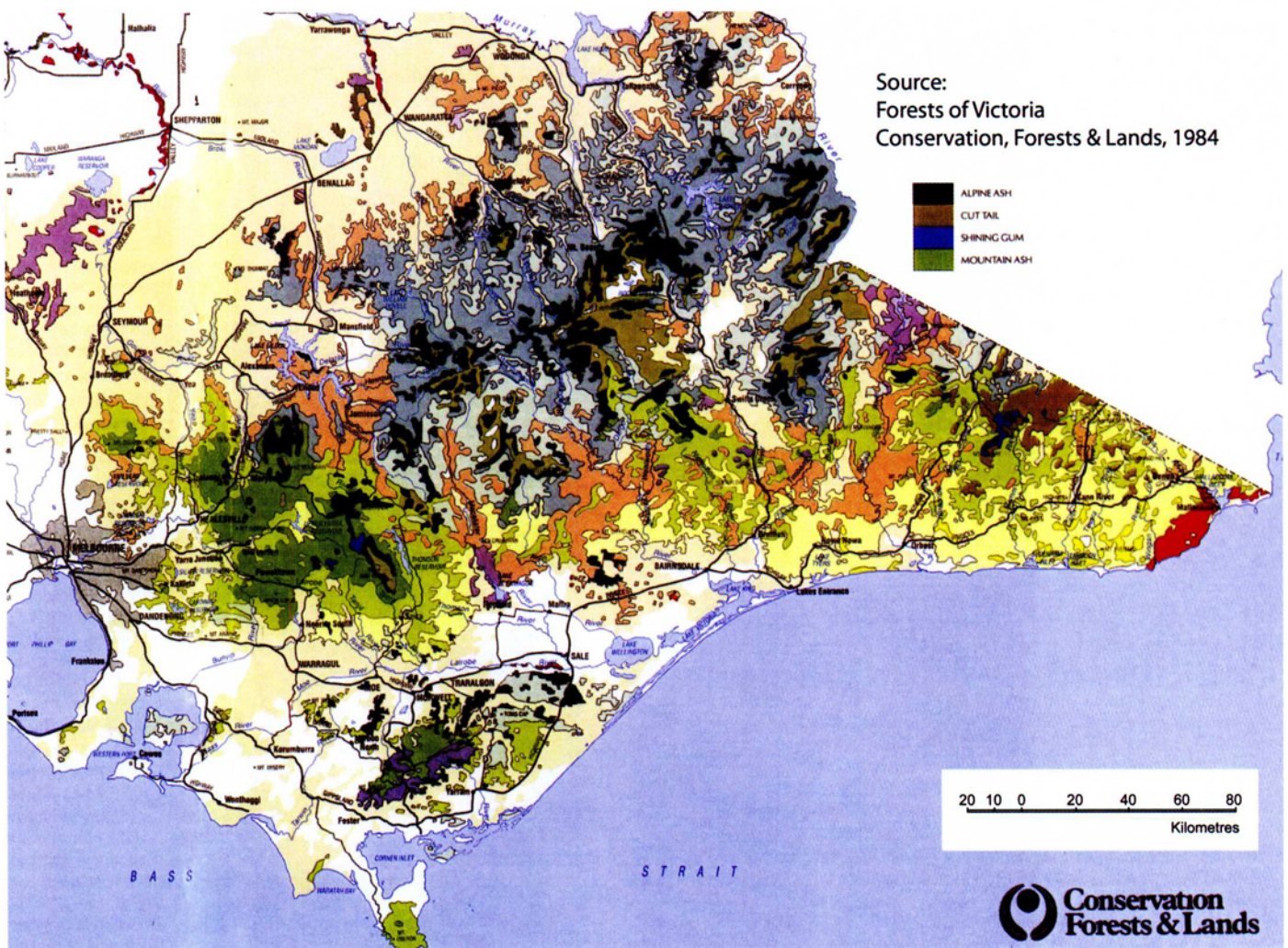


Figure 1. Location of ash-type forests in eastern Victoria

Many of the alpine ash state forests were also burnt and regenerated in 1939, but they have a wider geographic dispersal and more varied fire history, not least through successive fires in 2003 and 2006–2007 as well as 2009. Figure 2 is a composite picture of fire boundaries for the Central Highlands (2009), Great Divide (2006–2007) and Alpine (2003) fires.

Figure 2 does not distinguish between forest types, but the areas are so widespread and overlapping of those in Figure 1 that the magnitude of the impact on both the forest types under examination will be obvious.

Fire occurrence and extent

Ash-type forests are moist forest types, generally found at medium to high elevations, with relatively high rainfall, deep soil and, droughts permitting, abundant soil water reserves. Under most conditions they are seldom dry enough to support an intense fire. This makes fuel reduction burning impractical in all but the mixed forest margins or ecotone sites. Burning of the ash forests is in any event undesirable because of their sensitivity to fire and widespread mortality following fire. Under conditions of prolonged drought, however, they carry heavy loads of ground-based and elevated fuels, capable of supporting fires of high intensity. As Tolhurst and McCarthy (2004) have pointed out, fire intensities and rates of spread are more weather-dependent than fuel-dependent under extreme conditions, such as those above a Forest Fire Danger Index of 50.

Before the 1980s, our knowledge of the frequency and extent of past fires in these forest types was poor. Most of the prior descriptions were unable to differentiate forest types and estimates of areas burnt were imprecise. However, where the 1919, 1926 or

1939 fires killed extensive areas of trees, the resultant even-aged regeneration provided useful information that has been used in the estimation of the areas of age classes (Ferguson 2010, Table 1).

The issue of fire frequency, both in history and in the future, is important given the projections of future climate change. Past research is scanty and has highlighted the uncertain state of our knowledge of past fire frequency, let alone the issue of climate change.

The mean intervals between fires were used to characterise the probability distributions in Monte Carlo processes used to model the occurrence of fires in state forests. Because the estimates of the mean intervals between fires for mountain ash spanned such a wide range, a range of values was used in this study based on discussion with field observers. For mountain ash, values of 100, 80 and 60 y were used. For alpine ash, the corresponding estimates were 60, 50 and 40 y.

A trinomial probability distribution was used to differentiate between the years in which no fires occurred, those in which minor fires less than 2000 ha occurred, and those in which major fires larger than 2000 ha occurred. As with all probability distributions discussed in this study, the trinomial refers to the parent distribution. It is applied in an inverse cumulative form in Monte Carlo processes to enable a random value of the variable in question to be drawn. That value becomes a sample observation that is combined with those for other variables drawn similarly, enabling a simulation of successive years up to the planning horizon in a single run. The process is repeated many times, in this case for 2000 runs, for each set of decision or key variables, enabling probability distributions of the final economic and risk criteria of interest to be estimated and assessed.

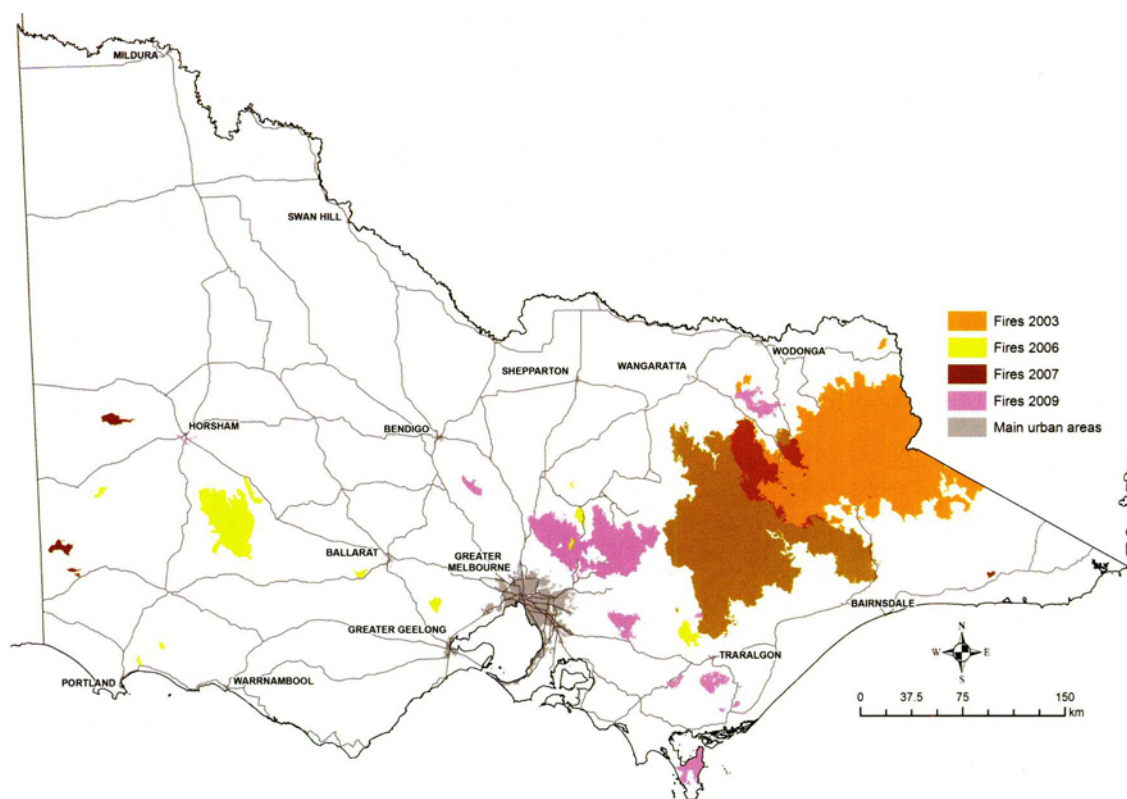


Figure 2. Areas burnt in 2003, 2006–2007 and 2009 fires in Eastern Victoria (Source A. Haywood, DSE)

The areal extent of minor fires was estimated using a truncated (at about 2000 ha) negative exponential probability distribution. That for major fires was estimated from a uniform probability distribution ranging from 2000 ha to the total area carrying sufficient fuel.

Climate change

But what of climate change? Nitschke and Hickey (2007) initiated a study of the implications of climate change for the future distribution of forest types. However, so far this research has not included a detailed analysis of the impact of disturbances such as fire.

Fortunately, Lucas *et al.* (2007) provided a historical analysis of past trends in the Forest Fire Danger Index (among others) over the period 1973–2007, together with projections to 2050. This index is a composite of rainfall and the time since rain, air temperature, relative humidity, wind speed and drought index. It was originally calibrated to a maximum of 100, corresponding notionally to general conditions of the Black Friday fire in 1939. Any value over 50 is said to constitute extreme forest fire danger and over 75, catastrophic danger.

Notwithstanding the obvious limitations of any such index, the results of Lucas *et al.* (2007) provided a means of taking climate change into account using the change in the number of days of extreme and catastrophic levels to 2050. The ratio of the values at the beginning and end of this period was 1.2 for their lowest projected rate of climate change (corresponding to an expected increase in mean temperature of around 1°C), while that for their highest rate of climate change was 2.32 (corresponding to an increase of about 2.5°C). If the mean intervals chosen earlier are divided by these values to obtain new estimates of the mean

intervals in 2050, they show a progression that looks sensible in terms of the likely impact of climate change on fire occurrence.

The mean intervals between fires were adjusted on a linear basis to account for climate change, such that by 2050 they were reduced by dividing the mean interval in 2009 by 1.2 for the lowest projected rate of climate change and 2.32 for the highest. No climate change can be represented by a ratio of 1.0 throughout.

Seed production

Both mountain and alpine ash are cyclical in seed production with marked peaks — on average every 4 y in mountain ash and 5 y in alpine ash. Commercial seed collection is feasible only when there is a sufficiently heavy crop of capsules, because trees have to be climbed and the branches cut, involving both skill and safety issues, as Figure 3 shows.

Using the data for mountain ash as a basis, and recognising that regional estimates across the entire forest type would tend to moderate the fluctuations apparent in that data, a stochastic model of the occurrence of good seed production years was formulated, involving an immediate decline after the peak and progressive increase to the next peak year. Finally, a stochastic overlay was imposed on those trends. Samples of the simulated time patterns of the proportions by area of natural regeneration are shown in Figure 4.

Simulation of seed collection was related to the proportion of natural regeneration, assuming that this proportion was also a reflection of the amount of seed held, on average, in capsules in the remaining areas of living trees over 29 y of age. The relationship was determined using a non-linear function that took the target level of seed collection for a ‘good’ seed production year as an upper limit. A stochastic component was also added to moderate the exact relationship. Figure 5 shows the simulated time patterns of seed production that were derived by applying this process to the data in Figure 4.

Economic criterion and risks

The discounted value of net costs was the economic criterion used to examine alternative seed supply storage and collection



Figure 3. Seed collection (Source: Justin McManus, FairfaxPhotos)

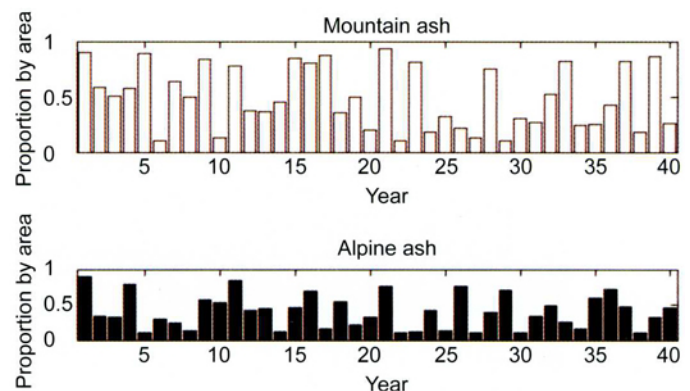


Figure 4. Sample of simulated proportions of area naturally regenerating after major fires

facilities, but this criterion and decision is not germane to this discussion. However, possible trade-offs between the costs and risks involved also had to be considered and the risks are very much germane to this study.

One undesirable risk was the biological criterion where, put simply, there was not enough forest of seed-bearing age plus seed in storage in 2050 to supply the average future demand for artificial regeneration over the ensuing 20 years.

Another risk was the timber production criterion, where the area that had reached rotation age or greater in 2050 was insufficient to maintain the harvest level over the ensuing 20 years.

Figure 6 is based on the results of two simulation runs deliberately selected to illustrate the risks visually. The upper panes of Figure 6 show a low-risk situation for seed production in 2050, because the age-class distributions for the two forest types are well spread and relatively even. The lower panes of Figure 6 illustrate a patently high risk, because the overwhelming majority of the area of the forest type is under 40 y of age and most of it is under 20 y in the case of alpine ash.

These two criteria can be defined quantitatively in terms of the variables available in the simulations and thus it is possible to summarise the outcomes in terms of probabilities over each set of 2000 simulation runs. Undesirable risks were then summarised

across all runs of the particular simulation to provide a probability criterion for assessing unacceptable risks. For example, if, taken over 2000 runs, 60% of the outcomes for a particular criterion were undesirable, the overall outcome might be deemed to constitute an unacceptable risk.

Key variables

The simulation study also had to account for several other variables besides mean intervals, climate change, seed-production years and extent of natural regeneration after major fires for each forest type, and some deterministic constraints. Other stochastic variables included the years of occurrence of minor and major fires respectively and the areal extent of those fires. The deterministic constraints included the allocation of burnt areas to age classes, confining seed collection to areas over 20 y of age, and restricting successive fires to areas of forest of at least 4 y of age that have sufficient fuel to burn.

Three key variables seemed likely to affect the outcomes materially, so a range of values was tested to explore the sensitivity of the outcomes. These were:

- seed collection and storage targets
- mean interval between fires
- climate change.

The major decision variable was obviously the targets set for seed collection in good seed production years, the other two being outside the control of DSE.

Results

State forests

Because the results from varying mean interval parallel those for varying climate change, attention can be restricted to the latter in this account¹.

¹ Attendees at the ANU Westoby Lecture in Canberra on 26 August 2009 may notice differences between Figures 7 and 8 and their analogues in that presentation. This reflects a subsequent inspiration to model the proportion of natural regeneration and seed collection as stochastic variables, rather than deterministic.

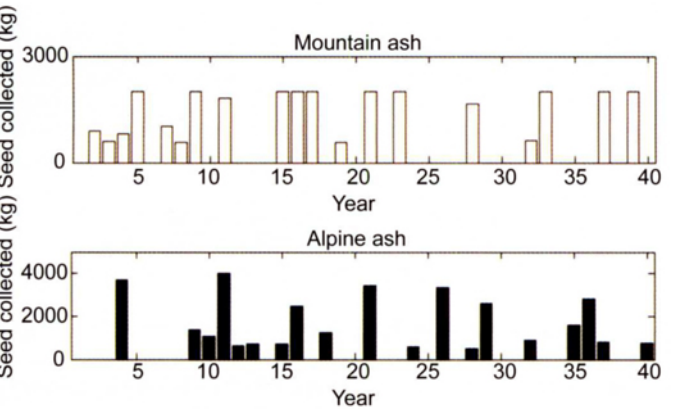


Figure 5. Simulation of seed collection based on data in Figure 4

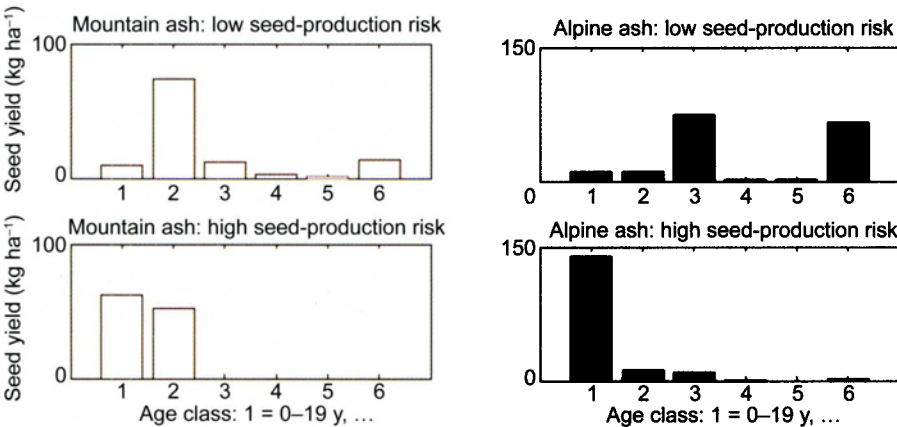


Figure 6. Examples of simulations showing low and high seed-production risks, based on age class distributions in 2050

A probability of 0.1 represents a risk of violating the criterion of one in ten, which is probably tolerable. A probability of 0.6 represents a risk of six in ten, which is not. More importantly for this analysis, an incremental increase in probability of 0.2 or greater constitutes a material unacceptable change.

Figures 7 and 8 summarise the results for the probabilities of unacceptable risks for two scenarios of Low and Median targets for seed collection in good seed-production years. The results for the High scenario target are not shown because they are only slightly less than those for the Median target and a High target is probably unattainable or inordinately costly in practical terms.

Figure 7 shows that, for mountain ash, increasing seed collection targets from 1000 to 2000 kg in good seed production years reduced the 2050 risk of having insufficient area of seed-producing age plus seed in storage to meet average artificial regeneration needs in the ensuing 20 y. However, this level of seed collection did not eliminate the risk, which is at the borderline of materiality if the rate of climate change is high.

Alpine ash, on the other hand, is at material risk (i.e. above a probability of 0.2) by 2050 if seed collection is low (2000 kg in a good seed production year), and at perilous risk (probability above 0.9) if climate change is low or greater. Increasing seed collection to the median 4000-kg level reduces the risk for all rates of climate change but there is nonetheless a material risk for a high rate of climate change.

Figure 8 shows that the viability of timber production in 2050 both forest types is not greatly affected by the target level of seed production. Both target levels show material and increasing risk as the rate of climate change increases, the risk levels for the two forest types under high rates of climate change and median seed collection levels being uncomfortably high (above 0.4).

Like all simulations, the results are open to interpretation based on views about various assumptions, probabilities and models involved. The actual outcomes could be worse if good seed production years are less frequent and fires more frequent and or

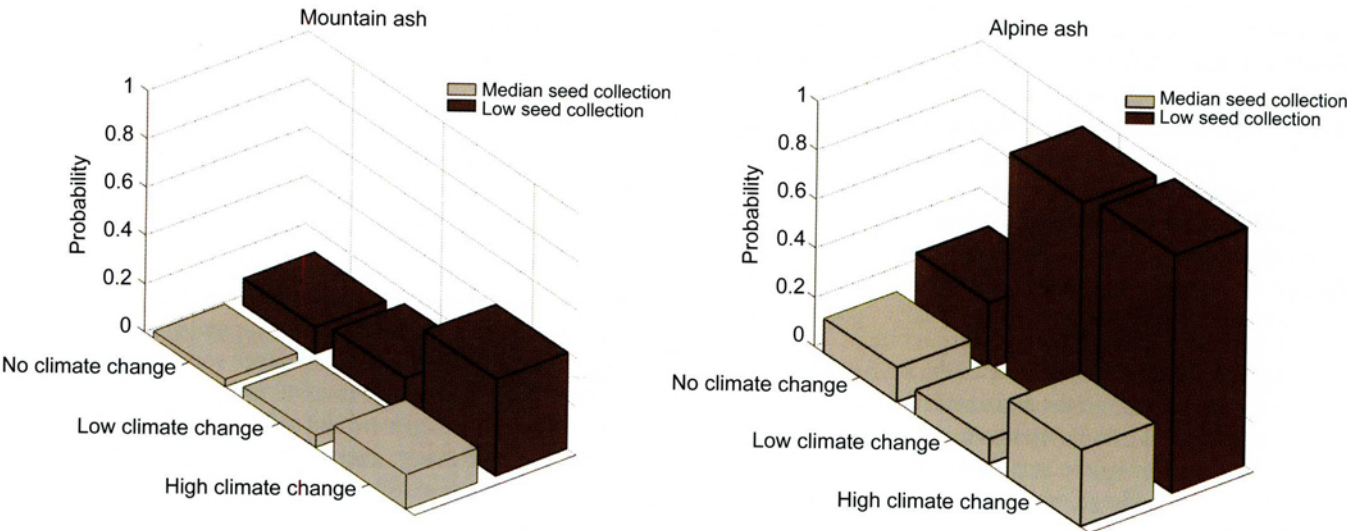


Figure 7. Seed production risks

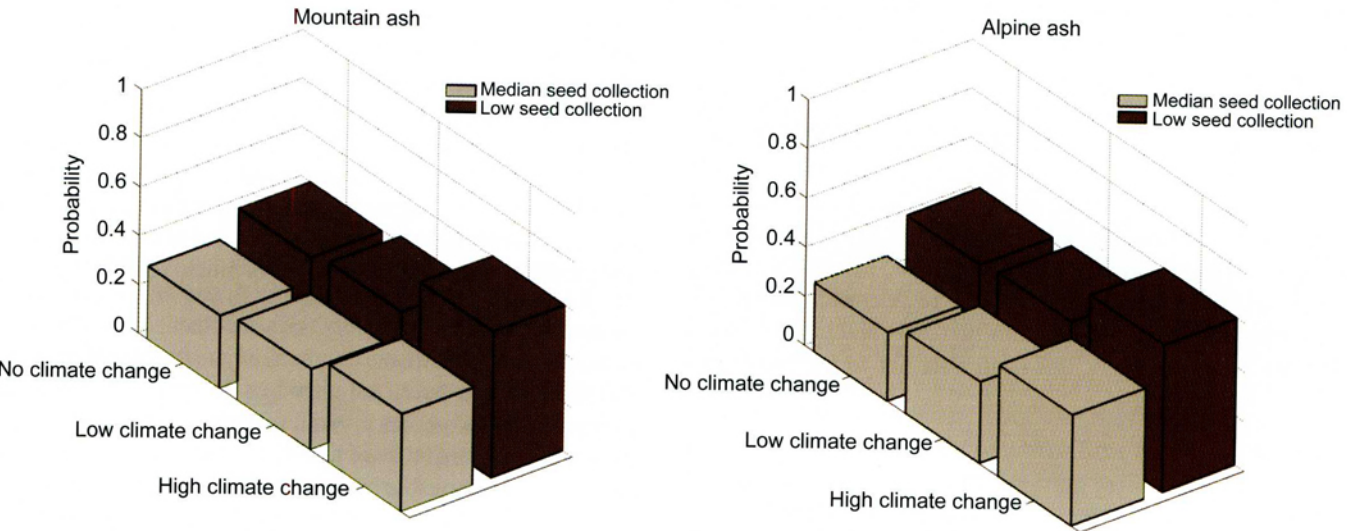


Figure 8. Timber production risks

extensive. Likewise they could be better if the converse holds. We have no way of knowing how the cards will fall. Quantitatively expressed risks provide the most appropriate guidance. The linkage of mean interval with climate change enabled a relatively simple stochastic analysis and a simple exposition of results based on a range of values for each, rather than the so-called 'info-gap' approach (see McCarthy and Lindenmayer 2007).

Conservation reserves

Some will ask, 'What about the conservation reserves?' It would be interesting to run a similar exercise but the data were not available at the time of undertaking this study. Present Parks Victoria policy is to not allow intervention through artificial natural regeneration and this greatly increases the risk to the survival of these forest types.

The mountain ash conservation reserves are mainly associated with Melbourne's closed water catchments and many carried iconic old-growth forests prior to 2009. In addition to their biodiversity values, these old-growth forests played an important role in increasing domestic water yields and decreasing turbidity. The O'Shannassy, Wallaby Creek², Maroondah, Bunyip and Tarago catchments carried old-growth mountain ash not previously burnt severely for over 150–250 y. Significant areas were burnt severely in 2009 (Melbourne Water 2009), early reports indicating that 93, 100, 76, 100 and 100% respectively of those catchments were burnt and much of the mountain ash forest killed. These forests will probably regenerate naturally given that 2009 was a good seed production year, but another fire in the next 20 y might be less kind, so the future risk to these forests is material, especially until the regenerating stands set seed.

The succession of fires in 2003 and 2006–2007 has also created an age-class distribution in the alpine ash conservation reserves that puts them at risk, probably much more so than the results reported for state forests, because artificial regeneration is not currently allowed and large areas are less than 20 y old.

What are the implications if regenerating stands of mountain ash and alpine ash are burnt prior to reaching 20 y of age and those species do not subsequently regenerate in the conservation reserves? There is, of course, great variation in the plant species composition in the ash-type forests, making detailed generalisations about future composition difficult. Much also depends on the subsequent fire frequency (Florence 1996). But the general tendency would be to promote more open, less rich and more fire-prone forest types with different animal and plant biodiversity and less attractive properties for domestic water supply, especially in relation to turbidity. Such changes are not an argument for artificial regeneration of ash-type forests across all conservation reserves, but there may be a cogent argument, especially relating to domestic water supply, for artificial regeneration on specific areas. Provided regeneration from the 2009 and earlier fires is adequate, as seems likely, this is an issue for future rather than

current policy. Protecting that regeneration for the next 20 y, however, is a matter that merits urgent consideration.

Active management

Franklin (2002) and others have stressed the need for *active management* of temperate natural forests, both conservation reserves and state forests. They are not referring to an open slather of exploitation but to a limited and conservative application of a range of activities that includes some salvage logging (Lindenmayer *et al.* 2008), some fuel reduction measures, some thinning of young stands to accelerate the development of structures characteristic of late succession, and measures to reduce the risks posed by exotic pests and diseases under climate change.

The National Conservation Reserve system is a very important achievement of the Regional Forest Agreement process and a much-needed component of management of Australian native forests. But the ash-type forests in conservation reserves (mainly national parks) are at least as much at risk as those in state forests, as the 2009 fires have shown. Hence the need to consider active management measures that cross over the public perception of the tenure boundaries. Ash-type forests, whether state forest or conservation reserve, have especially important roles for a wide range of uses. If we continue to neglect the risks fire poses, we may not be able to sustain the uses and values we currently use or ascribe to them.

The question, then, is how to effectively manage such risks. Seed collection for artificial regeneration of areas burnt but that are unable to regenerate naturally are only one part of the solution. While the past reluctance (if not legal inability) of the managers of conservation reserves to countenance some of these measures in national parks and related reserves is entirely understandable, a major review of management and policy is needed. Like Franklin, Lindenmayer and many other ecologists and foresters, I believe active management of ash-type forests is essential and needs to include:

1. A well-equipped workforce

While helicopter and fixed-wing water tankers and helicopter crews are extremely useful for first attack on fires, they generally have to be backed by the construction or renovation of tracks and associated measures to contain the fire on the ground, using vehicular water-tankers and trained personnel. DSE mostly contracts in bulldozers to assist and likewise Parks Victoria and Melbourne Water, who contribute to fire suppression on Melbourne's water supply catchments in national parks. But all are limited in the trained staff available and in the bulldozers and other resources that can be brought in quickly for first attack.

Timber harvesting contractors form a potentially important and somewhat under-utilised component in this workforce in an otherwise very sparsely populated environment, in which local knowledge and experience in tall forest is critical. Swing blades and water tanks can be fitted to wheeled skidders and forwarders, possibly on a removable basis, as Jaffe and O'Brien (2009) have shown. The formation of more mobile harvesting crews, both forwarders and tree-fallers, could assist timely first attack, as

² Ashton (2000), a much respected ecologist, provided a detailed account of 'The Big Ash' in Wallaby Creek in 1949 compared with 1997. Sadly, his closing words in the abstract — 'Whether the Big Ash will be spared from fire in future centuries is very doubtful' — proved all too accurate.

well as later containment of major fires in moderate topography, following the cessation of extreme weather conditions. Attention and possibly subsidies need to be given to the design and use of harvesters that are better-suited to dual usage for harvesting and fire fighting, with as small a soil disturbance footprint as possible.

Few harvesting contractors have large bulldozers, so fire-management agencies will continue to draw mainly on earth-moving contractors for these. In common with all vehicles used in fire suppression, these need to be fitted with automated vehicle location systems for personnel safety and identification of containment boundaries.

2. Removal and salvage of fire-killed trees along access roads

Fire-killed trees along roads and access tracks later become hazards in terms of torching and spotting in later fires, or impeding access when they fall over roads or access tracks. They should be felled and where possible sold for timber, substituting for harvest volume from living forest. Agencies presently fell trees that are dangerous along tourist roads, move the trunks away from the road where they impede access and constitute additional fuel that is difficult to extinguish, and even mechanically slash under the residual trees in or adjacent to some conservation reserves. Utilisation of sound timber hardly has a devastating impact under these circumstances.

Similarly, felling of dead trees on ridges may be necessary to reduce potential spotting if the subsequent regrowth is ignited. Likewise felling of fire-killed trees on aesthetically important landscape units may be required.

Salvage logging³ needs to be subject to transparent and timely due process, based on some well-established protocols (Lindenmayer *et al.* 2008). For example, Lindenmayer and Ough (2006) have mounted a persuasive argument that salvage logging of fire-killed old growth ash-type forest should not be permitted, and the protocols should reflect this argument, except in prescribed exceptional circumstances.

3. Seed collection and artificial regeneration in conservation reserves

Seed collection is a strategically vital matter if the risks posed by fire are to be offset. Access to conservation reserves for seed collection is currently restricted except for purely botanical studies. Greater access needs to be accorded to seed collection, given the magnitude of the task.

Biodiversity values are as much at risk as others, so the general stricture against the use of artificial regeneration also needs to be reconsidered, especially on water catchments where that use has at least as great a priority as maintaining biodiversity through natural processes. This is not an argument for universal use of artificial regeneration in the ash-type forests but rather a limited and strategic program using sensible protocols and mechanisms.

4. Strategic swaps of age classes to the advantage of both tenures

Marginal swaps of regrowth areas between the conservation reserves and state forests can potentially reduce the risks to both. Recruiting some older regrowth into the conservation reserve in exchange for salvage of fire-killed regrowth can reduce the totality of harvest removals plus mortality to the advantage of the future timber production, carbon footprint, biodiversity and water production. Adding some older regrowth to conservation reserves can provide a more robust distribution of age-classes, successional stages and biodiversity. This is part of broader philosophy of looking at vegetation types across the landscape in a holistic way and creatively considering the best way to maintain patterns of age-distributions and ecosystem resilience, irrespective of tenure.

This would require legislative changes to permit timely resolution with respect to salvage logging and associated protocols to protect both forms of tenure from exploitation.

5. Regrowth thinning for habitat management and water production

Lindenmayer (2002) has made the case that some thinning of regrowth stands may be desirable in order to maintain populations of Leadbeater's possum. Given the extent of the recent fires in state forests and conservation reserves, the situation warrants review in order to achieve the best outcomes. Similar legislative and related changes to those outlined in the previous point may be needed.

Thinning of mountain ash regrowth stands to increase water production was the subject of earlier research (O'Shaughnessy *et al.* 1993). The experiments are being remeasured and assessed but only preliminary results are available to date (Hawthorne 2008). Because of the preponderance of mountain ash in Melbourne's water catchments, thinning to increase water production in conservation reserves warrants consideration in any new protocols.

6. Better access for fire management and more fuel reduction burning

Further investment in the provision of access tracks and the reduction of hazards along them by mechanical slashing is needed. DSE had commenced such a program prior to the 2009 fires — a move I viewed with some scepticism prior to the 2009 fire, but no longer. If the risks to the ash-type forests are to be reduced, better access in state forests and conservation reserves is essential. Maintenance of all roads and tracks in or serving the ash-type forests is equally important: it is a practice that often tends to slip when budgets are tight. These measures will not stop a head fire burning under extreme conditions but they may assist containment of flank fires and assist in attacking head fires when conditions moderate. They also may reduce the likelihood of linking of nearby lightning strike fires and hence reduce the overall potential for a major fire front to develop.

More strategic fuel-reduction burning of peripheral mixed forest types is needed. DSE has already started to implement an

³ DSE has guidelines for salvage logging but these do not extend to many of the measures advocated here.

increased program of landscape-level fuel-reduction burning. However, the risks posed to the future of the ash-type forests call for a strategic review of the program, specifically to protect those forests.

A fruitless debate has raged over fuel reduction burning, especially in relation to the frequency, intensity and patchiness appropriate to landscape-level burning. Gill's (1999) dictum amounts to varying the timing, frequency, intensity and patchiness as much as possible. But as Clarke (2008) has pointed out, pleas for burning to achieve a mosaic lack practical criteria for implementation. As with a number of the other active management measures, monitoring of and research on the biodiversity, timber and water production impacts will be needed to refine the protocols and mechanisms — in some cases requiring major multi-disciplinary research projects.

7. Strategic orientation of harvest areas

Contrary to Gill's (1999) view, there is ample anecdotal evidence that young (under 5 y) regrowth resulting from harvesting, together with the harvest road system, can assist suppression and containment, except under extreme conditions. While further study is warranted, this opens up the issue of strategic location of harvesting to protect key adjacent conservation assets.

Future policy

Let me now summarise some of the implications for forest policy. In the past, the conservation versus development debate about forests has been a 'zero-sum game'. Whatever one side gained the other lost, as areas were transferred from one designation to the other.

That approach neglects the role of a third player, Mother Nature, especially relating to fire. We need to move away from the 'one tenure – one use' thinking (Ferguson 2005) that underlies the zero-sum game and recognise that there are multiple uses and values associated with the ash-type forests, many of which have no special relationship with the nominal tenure.

The ash-type conservation reserves are as much about multiple uses as are state forests. Some uses are common to both, some are excluded from one and not the other, and some vary by degree rather than absolute exclusion in one or the other. If we want to maintain the ash-type forests, we need *active management* that integrates across those tenure boundaries (Ferguson 2007) especially in relation to protecting against the risks posed by fire, the biggest threat. Without radically altering present tenures or principles of management, co-operative arrangements and trade-offs at the margin would enable both tenures to benefit, as well as public forest ownership in general. Transparent and timely processes and clear protocols are essential. Protocols will need to be reviewed in the light of further monitoring and research, which need to be built into ongoing budget provisions. Without better information, public interest and support for these measures will decline as memories of the recent fires fade.

There will be some who find some of the measures suggested distasteful, or worse. Yet we already countenance the culling of kangaroos in conservation reserves in order to protect the

populations and their habitat, notwithstanding the understandable distaste that some in the community have for the practice. We sterilise koalas in conservation reserves and elsewhere for similar reasons and with similar concerns. And we remove dangerous limbs, fell trees in picnic and parking areas, fell dangerous trees along roadsides and even sometimes mechanically slash under them in conservation reserves and related areas. The purist rhetoric of one tenure – one use that is often featured in the media is a false dichotomy that ignores the multiple uses and tradeoffs involved and the stochastic variability of natural processes, and promotes an unrealistic sylvan fundamentalism in the public domain.

Jack Westoby and Alf Leslie were very much of the view that managing forests was primarily about managing them for people, both now and in the future. If we are to maintain the values of these iconic forests for future generations, in a future climate that may be quite different from the current one, we will need to pursue active management, based on monitoring and research. Whatever ethical or philosophical position one might take on discount rates, this will involve costs and immediate and ongoing actions.

We face a future in which there is a material risk we could lose most or all these ash-type forests, not over a period of several millennia, such as that in which they evolved, but in the next 60 y. We can decide to do nothing and accept that loss, or we can engage in active management to conserve their values and uses.

In posing that choice, I hear Alf Leslie reminding me 'that may be a cogent and compelling case you have argued, but don't forget, when it comes to public policy, stupidity nearly always wins' (Leslie 2000). That said, I know that Alf loved the ash forests and would be delighted to be proven wrong in this case.

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